

Technical Document #426: Natural Language Processing - Comprehensive Overview

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Tokenization is the process of breaking text into individual units called tokens. Subword tokenization methods like BPE handle out-of-vocabulary words gracefully.

Language models predict the probability of a sequence of words. Large language models like GPT and BERT have achieved remarkable performance on various NLP tasks.

Coreference resolution identifies all expressions that refer to the same entity in a text. It is essential for understanding document-level coherence.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Question answering systems extract answers from a given context.
- Word embeddings represent words as dense vectors in continuous space.
- Coreference resolution identifies all expressions that refer to the same entity in a text.